

## PRODUCT DESCRIPTION

Master Seal is a highly efficient hydrophobising and water-repellent powder additive for cementitious systems, supplied ready to use with no dilution. It is blended into the dry mix of renders, plasters, masonry mortars, concrete and tile jointing mortars, where it renders the internal pore walls of the hardened material water-repellent — cutting capillary water uptake through the full body of the material, not at the surface only. Water beads on the face instead of being drawn into the pores, the surface dries out faster after rain, and the conditions that lead to dark staining and biological growth are markedly reduced. Dosage response is linear, so the level of repellency can be set predictably. It does not retard setting, does not foam, and does not reduce surface hardness, adhesive strength or compressive strength. Raw material: premium grade, imported from Switzerland.

## KEY DATA

|                            |                                                                                                                                                                         |
|----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Type                       | Water-repellent, bulk-hydrophobising powder additive for cement-based mortars, renders, concrete and jointing compounds                                                 |
| Raw material               | Premium grade, imported from Switzerland — supplied undiluted                                                                                                           |
| Appearance                 | Free-flowing white powder, non-dusting, odourless                                                                                                                       |
| Bulk density               | 650–850 g/L                                                                                                                                                             |
| Residual moisture          | < 2.0 %                                                                                                                                                                 |
| pH value                   | 9.0–11.0 (10 % dispersion in water)                                                                                                                                     |
| Alkali stability           | Fully effective at typical cementitious pH 11–12                                                                                                                        |
| Dosage — mortar / concrete | 0.2–0.5 % by weight of dry binder for general work (50 kg cement » 100 g at 0.2 %); up to 1 % where a high degree of repellency is required — linear response           |
| Dosage — tile grout        | 2 % by weight of dry jointing mortar (1 kg grout » 20 g) — the higher rate reflects the thin section and permanent wetting of a grout joint                             |
| Suitable systems           | Plasters and renders, masonry and rendering mortars, decorative renders, concrete, cementitious tile adhesives, tile jointing compounds, and sealing mortars / slurries |
| Workability                | Excellent and neutral — unchanged even at elevated dosage rates                                                                                                         |
| Mixing                     | Blend into the dry mix before water is added; keep mixing time short; let the wetted mix stand 1–5 min, then stir briefly again                                         |
| Effect on the mix          | No retardation, no foaming; surface hardness, adhesive strength and compressive strength unchanged; water demand unchanged                                              |
| Compatibility              | Compatible with standard mortar additives — redispersible polymer powders, accelerators, retarders, cellulose ethers                                                    |
| Overcoating                | Normally paintable at 0.2–0.5 %; adhesion falls at higher dosages — trial first                                                                                         |
| Packaging                  | 100 g · 400 g · 800 g                                                                                                                                                   |

## MECHANISM

Master Seal acts chemically inside the pore structure of the hardened cementitious material. The internal walls of the capillary pore network are rendered water-repellent, so liquid water is no longer drawn in by capillary action; it beads on the face instead. Because the additive is distributed through the mix rather than applied to the surface, protection is present in the bulk of the material as well as at the face, and it is not lost when the surface is worn, cleaned or lightly abraded. The pore network itself remains open, so the material still breathes and dries out. This is a water-repellency mechanism, not a pore-blocking or crack-bridging one: it reduces absorption, it does not seal a defect.

## NOTE ON VALUES

Dosage figures relate to the weight of the dry cementitious binder, not to the mixing water. The repellency achieved depends on mix design, binder, aggregate grading, water/cement ratio, compaction and curing. Figures in this document are typical guidance only and are not a specification.

## APPLICATION NOTES

- Always blend the powder into the dry mix first, mixing until the dry blend is uniform in colour. Never add it to mortar that has already been mixed with water — it agglomerates and disperses unevenly, leaving treated and untreated zones in the same batch.
- Weigh the dosage; do not judge it by eye. Avoid prolonged high-speed agitation of the wetted mix: excessive shear entrains air, and entrained air voids become water paths in the hardened material.
- Factory dry-blending: Master Seal can be dry-blended with other dry additives in any standard compulsory mixer to produce bagged cement-based products. Keep mixing time short and avoid major temperature increases caused by high shear forces.
- Site mixing of a finished dry-mix product: place the required amount of mixing water in the vessel first, then add the powder blend into the water under agitation. Pouring water onto a heap of dry powder produces lumps that are difficult to break down.
- Do not alter the water demand of the mix and do not add other agents to assist the additive. Cure the mortar or concrete normally — the repellent effect becomes clearly visible once the surface has dried and cured.
- Where the surface is to be painted, keep the dosage within 0.2–0.5 % and always trial a small test patch before painting the whole area. At higher dosages, assume the surface will not be painted.
- Master Seal is not a waterproofing membrane: it reduces capillary absorption within the body of the material, does not bridge cracks and is not intended to resist continuous hydrostatic pressure. For swimming pools, water tanks, ponds and roof decks use it as a supplementary measure alongside a surface waterproofing system, and repair cracks and spalls in full before coating.
- Powder that has taken up moisture and caked will not disperse properly — discard it rather than crushing it into a batch.

## HEALTH & SAFETY

Master Seal is a dry, low-dusting, odourless powder. Handle it with the precautions that apply to any cement work: protective gloves, eye protection and a dust mask; avoid inhaling dust and contact with eyes and skin. Note that the cementitious binders into which it is mixed are themselves strongly alkaline and can burn skin and eyes on prolonged contact. Keep out of reach of children and animals. Refer to the Safety Data Sheet for full information.

## STORAGE

Store in the original sealed container in a dry place below 25 °C, away from direct sunlight. The powder is moisture-sensitive — reseal immediately after use; containers left open, stored damp or stacked under pressure tend to cake. Shelf life 6 months from delivery when stored as described.